

PROCESS

A Simple Guide to Separation Process Design, Cross-Flow Filtration, PFD/PCID Basics, and Scale-Up

A plain-language guide for anyone moving a membrane process from the lab bench to a working pilot plant.

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How to Use This Guide

This guide walks through four steps of turning a lab-scale membrane idea into a real, working process: designing the separation, running it on a pilot rig, drawing it up the way an engineer would, and scaling it up toward a commercial plant. The four sections build on each other in that order — what you decide in Section 1 shapes how you build and run the pilot in Section 2, which is what gets drawn up in Section 3, which becomes the evidence for the scale-up case in Section 4.

The word “membrane” is used here as a general term. Everything in this guide applies whether you are working with microfiltration (MF), ultrafiltration (UF), nanofiltration (NF), reverse osmosis (RO), or organic solvent nanofiltration (OSN, for solvents instead of water).

1. Separation Process Design

How to choose a process, pick a membrane, and work out the basic numbers before you build anything.

1.1 Choosing a Process and a Membrane

The first question is simple: how small is the thing you need to remove or keep? That decides which membrane family fits the job.

- Microfiltration (MF) — stops bacteria and bigger solid particles. Low pressure.
- Ultrafiltration (UF) — stops proteins and other large molecules. Still fairly low pressure.
- Nanofiltration (NF) — stops small molecules and most multi-charged ions, but lets some salt through.
- Reverse Osmosis (RO) — stops almost all dissolved salt. Needs the highest pressure of the four.
- Organic Solvent Nanofiltration (OSN) — same idea as NF, but built to survive solvents instead of water. Needs a solvent-resistant membrane material.

Beyond pore size, ask three more questions: Will the membrane survive the feed's pH, heat, and chemicals? How dirty or fouling-prone is the feed? And how much pressure (or other driving force) do you realistically have to work with? These three answers usually narrow the choice down fast.

1.2 Membrane Module Shapes

Once you know the membrane type, you still need to pick a physical shape for it. Each shape trades off how much membrane area you can fit in a given space against how easy it is to clean.

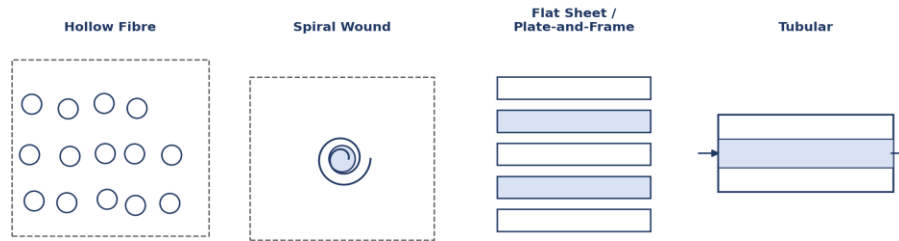


Figure 1.1 — The four most common membrane module shapes.

Shape	Area per Volume	Fouling Tolerance	Typical Use
Hollow fibre	Very high	Low–moderate	Clean feeds, gas separation, OSN
Spiral wound	High	Moderate	RO/NF desalination, most industrial UF/NF
Flat sheet / plate-and-frame	Low	High, easy to clean	Lab testing, very dirty feeds
Tubular	Low	Very high	Thick or chunky industrial feeds (dairy, slurries)

Table 1.1 — The four module shapes, compared.

1.3 Running Multiple Stages

One membrane stage almost never gets you all the way to the recovery you need. The usual fix is to send the leftover (retentate) from one stage into a second stage, and so on, so each stage only has to do part of the job. The cleaned-up streams (permeates) from every stage are usually combined into one outlet.

Two-Stage Membrane Cascade with Combined Permeate Header

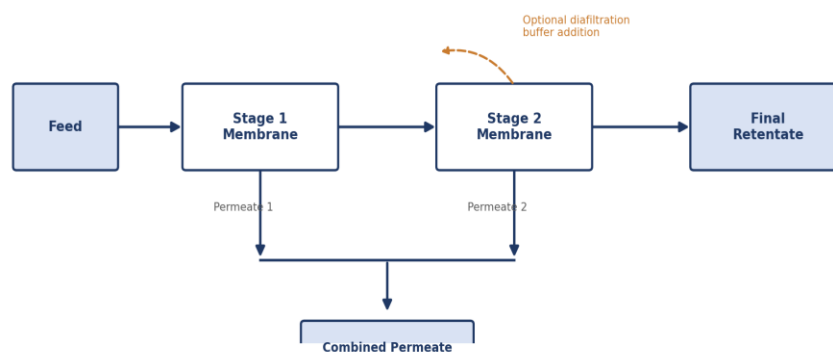


Figure 1.2 — A two-stage setup with one combined permeate outlet.

There are three basic ways to run a membrane system, and almost every process uses one of them:

- Batch (total recycle) — nothing leaves; useful for short test runs.
- Concentration mode — the permeate leaves, the tank gets smaller and more concentrated over time.
- Diafiltration — fresh liquid is added at the same rate permeate leaves, so the tank stays the same size while small unwanted molecules wash out.

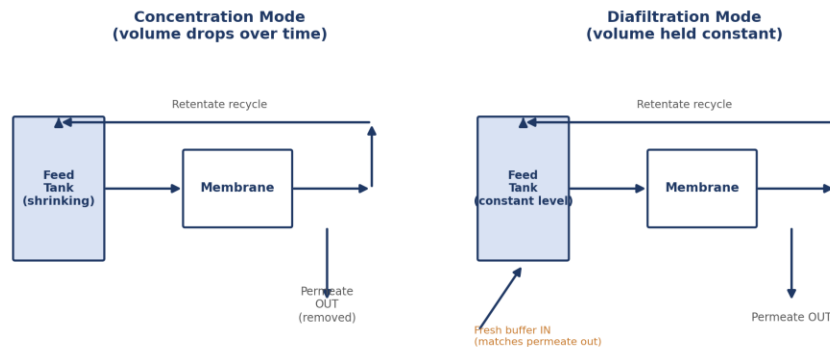


Figure 1.3 — Concentration mode (tank shrinks) versus diafiltration mode (tank stays full).

1.4 The Basic Numbers (Mass Balance)

Everything else in this guide comes back to a few simple equations. For one stage, with feed flow Q_f , permeate flow Q_p , and retentate flow Q_r :

$$Q_f = Q_p + Q_r \quad C_f \cdot Q_f = C_p \cdot Q_p + C_r \cdot Q_r$$

From this, two numbers tell you almost everything about how well a stage is doing:

$$\text{Recovery (Y)} = Q_p / Q_f$$

$$\text{Rejection (R)} = 1 - (C_p / C_f)$$

This rejection number is what you actually measure, so it is always a little lower than the membrane's “true” rejection — right at the membrane surface, the feed is always a bit more concentrated than in the bulk tank. For a batch that is simply being concentrated, the concentration factor (CF) tells you how much smaller the volume has gotten:

$$CF = V_{f,\text{initial}} / V_r \quad C_r \approx C_{f,\text{initial}} \times CF^R$$

For diafiltration, this equation tells you how fast an unwanted small molecule washes out as you add diavolumes (N) of fresh buffer:

$$C / C_0 = \exp(-N \times (1 - R))$$

Do the simple check often: feed in should equal permeate out plus retentate out, within a small error. If it doesn't, trust your flow meters and samples less, not your process.

1.5 Finding the Operating Window

The “operating window” is just the safe zone of pressure, flow, and concentration where the process runs well without fouling too fast. You build it in three simple steps:

1. Raise the pressure step by step and watch the flux. At first flux rises with pressure. Past a certain point it flattens out — pushing harder past that point just wastes energy and fouls the membrane faster.
2. Run the batch down and watch how flux falls as the feed gets more concentrated. This usually sets the practical limit on how far you can concentrate.
3. Run at your chosen point for a while and track how fast flux drops, so you know how often to clean.

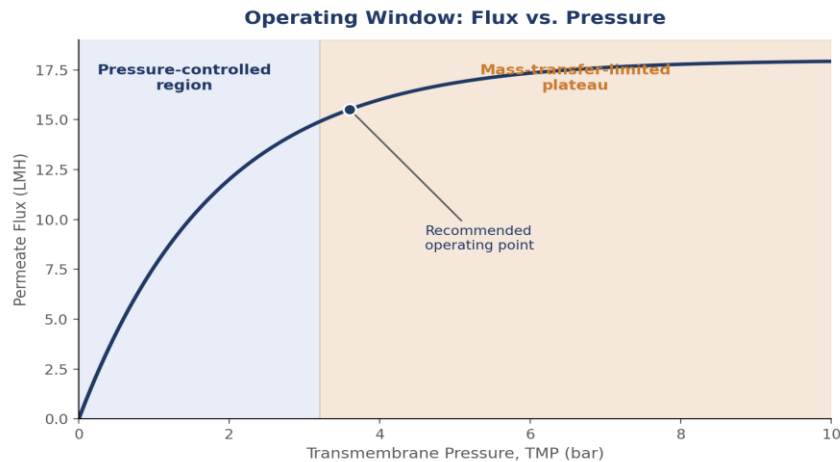


Figure 1.4 — Flux flattens out above a certain pressure. The best operating point sits right at that knee.

The result is usually one simple sentence, like: “run at 4–6 bar, 35–45°C, up to a concentration factor of 8, and clean every 12–24 hours.” That sentence becomes the design basis for Section 2.

1.6 Real Example: ExxonMobil's MAX-DEWAX® Process

Making lube oil involves chilling it with a solvent (MEK and toluene) to crystallize out the wax, then filtering the wax away. The hard part has always been recovering that solvent afterward, which traditionally meant boiling and re-condensing huge volumes — a slow, energy-hungry step.

Mobil (now ExxonMobil) and W. R. Grace built a membrane alternative instead: a polyimide membrane that lets the small solvent molecules through while blocking the much bigger oil and wax molecules. It is the reverse of how people usually picture a membrane working — a clean example of picking the right membrane for the actual

size gap in your stream, exactly as in Section 1.1. The first commercial unit started up at ExxonMobil's Beaumont, Texas refinery in May 1998, under the brand name MAX-DEWAX®.

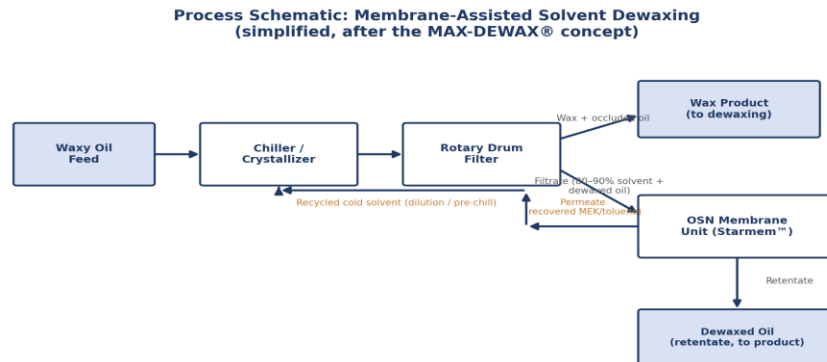


Figure 1.5 — A simplified schematic of the MAX-DEWAX process.

The real numbers make Sections 1.4 and 1.5 concrete: the Beaumont unit handles about 11,500 m³/day of product and recovers roughly 300,000 m³ of clean solvent every year. Recycling that solvent, instead of re-boiling it from scratch, cuts energy use by about 20% per unit of product, saves an estimated 36,000 barrels of fuel oil a year, and cuts solvent vapour losses by 50–200 tons a year. More than 25 years later, it is still the largest publicly documented industrial OSN installation in the world.

2. Cross-Flow Filtration

How a real cross-flow rig is laid out, controlled, watched, and tested.

2.1 How the System Is Laid Out

Cross-flow filtration pumps feed sideways across the membrane at speed, instead of straight into it. That sideways flow keeps sweeping the membrane surface clean instead of letting a thick layer build up. The basic parts are the same on almost any MF/UF/NF/RO/OSN rig:

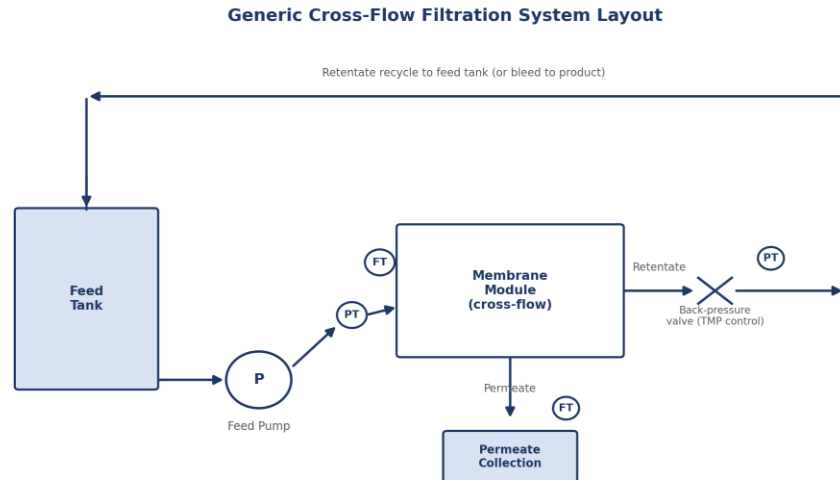


Figure 2.1 — A typical cross-flow filtration loop.

- Feed tank — holds at least one full system's worth of liquid, with some extra room.
- Pump — keeps the feed moving fast across the membrane.
- Membrane module(s) — sized using real pilot data, arranged using the staging logic from Section 1.3.
- Back-pressure valve — sets the pressure on the membrane separately from the pump speed.
- Sensors — at minimum, pressure, flow, and temperature, so you can see what's actually happening.

2.2 Controlling Pressure

The pressure that actually pushes liquid through the membrane is called transmembrane pressure (TMP). It is not just whatever the pump shows — it depends on both sides of the membrane:

$$\text{TMP} = (\text{Pf} + \text{Pr}) / 2 - \text{Pp}$$

In practice, you set TMP by adjusting the back-pressure valve, while the pump keeps the cross-flow speed steady. Keeping these two jobs separate — one valve for pressure, one pump for flow — is exactly why Figure 2.1 shows them as two different parts. Always try to run inside the pressure-controlled region from Figure 1.4. Running above it just burns extra energy.

2.3 Handling Permeate and Retentate

Where the permeate and retentate actually go decides which mode from Section 1.3 you are really running:

- Total recycle — everything goes back to the tank. Good for short test runs.
- Concentration mode — permeate leaves, tank shrinks.
- Diafiltration — permeate leaves, fresh liquid is added at the same rate.

One easy mistake to avoid: always take feed, permeate, and retentate samples at the same moment. Otherwise your mass balance from Section 1.4 won't actually add up, and you won't know whether that's a real process problem or just a sampling mismatch.

2.4 Watching Flux and Fouling

Flux — how much permeate you get per square metre of membrane per hour — is the single most useful number to watch in real time. A few simple derived numbers make it even more useful:

- Normalized flux — flux corrected for temperature, so a cold morning doesn't look like fouling.
- Flux decline rate — how fast flux drops at constant pressure. A sudden change in that rate usually means something changed (e.g., a cake layer just started forming).
- Permeability (flux $\div$ pressure) — strips pressure out of the picture, so you can tell a true fouling problem from a simple pressure change.

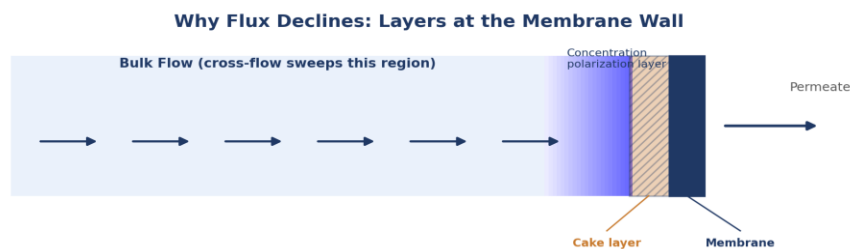


Figure 2.2 — Why flux drops: a thicker boundary layer, then a cake layer, sit between the bulk feed and the membrane.

2.5 Testing for Stability

Before trusting pilot results enough to scale up, prove the membrane can actually survive your real feed — don't just assume it:

4. Integrity test — check for leaks or oversized pores before you even start.
5. Clean-water flux, before and after — tells you whether any flux loss is reversible fouling or permanent damage.
6. Long soak test — run the real feed, at real temperature, for as long as (or longer than) your planned campaign. Many failures only show up after days, not hours.
7. Rejection re-check after the soak — if rejection drops, the membrane has been damaged or has degraded chemically.

For tough feeds — organic solvents, extreme pH — this testing is not optional. It is the only real evidence the membrane will survive at all.

2.6 Real Example: Whey Protein Concentrate Production

Cross-flow UF of cheese whey is one of the biggest, longest-running cross-flow membrane operations anywhere, and it shows almost every idea in this section at industrial scale. Whey contains valuable protein (about 0.6–1% by weight) along with lactose, fat, and minerals. UF concentrates and cleans up that protein into whey protein concentrate (WPC) or isolate (WPI).

Real industrial plants run continuously, with the largest using up to 12 membrane stages in a row — the same cascade idea as Figure 1.2, just at a bigger scale. Typical pressures run 2–5 bar, matching the kind of range Section 1.5 recommends finding for any new process. The first UF stage usually concentrates the whey about 4–5× by volume, and later stages push that further, up to 5–18×, to hit the final product spec.

Fouling is not a side issue here — it is the main thing operators manage every single shift. Whey's fat and minerals foul UF membranes fast enough that the resistance breakdown described in Section 2.4 (how much is reversible vs. permanent) is standard daily practice, used to decide exactly when to stop and clean.

3. PFD & P&ID Basics

How to read and draw the two main diagrams every real plant needs.

3.1 Process Flow Diagrams (PFDs)

A PFD shows the big picture: the major equipment, the streams connecting them, and the key numbers (flow, composition, temperature, pressure) — but none of the small valves or wiring. It's the right diagram to show a manager, a new hire, or a funder who wants to understand what the process actually does.

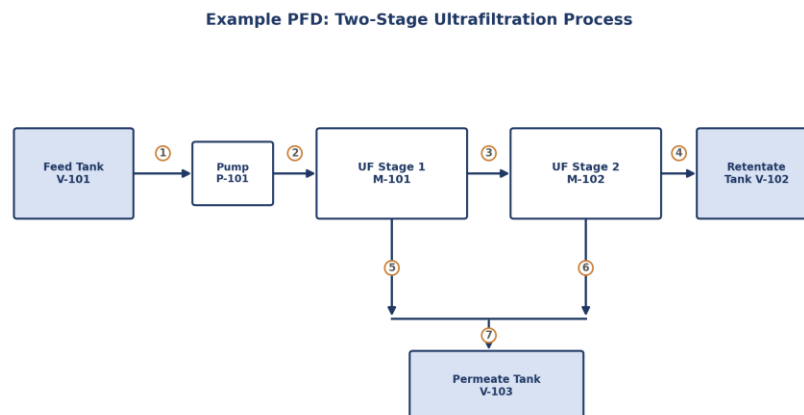


Figure 3.1 — An example PFD: a two-stage UF process, with each stream numbered.

Good habits for a PFD:

- Draw equipment as simple labelled boxes, not detailed drawings.
- Number every stream and list flow/composition/temperature/pressure for each one in a stream table.
- Always show recycle loops — they're often the most important part of the design.
- Leave out small valves and instruments. That's what the P&ID is for.

3.2 P&IDs and How Instruments Are Labelled

A P&ID adds everything the PFD leaves out: every pipe, valve, and instrument needed to actually build and run the plant safely. P&IDs use a standard set of symbols (based on a standard called ISA-5.1) and a standard naming system for instruments, so any trained engineer can read one without needing a legend.

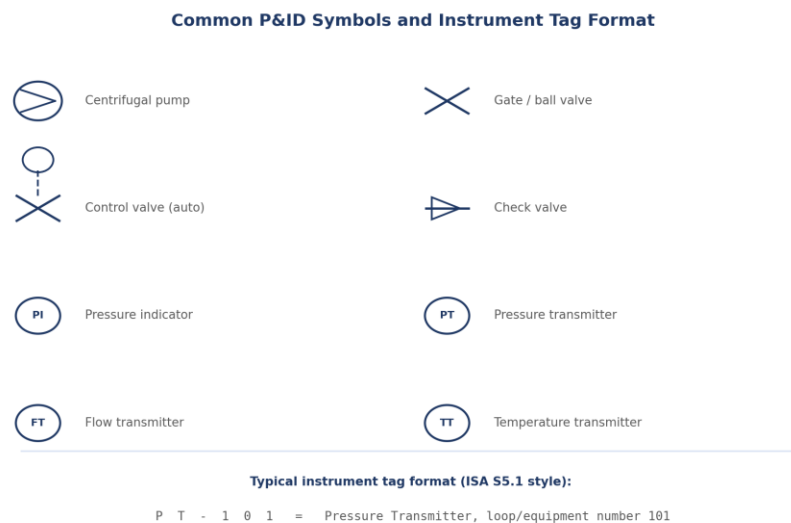


Figure 3.2 — A few of the most common P&ID symbols, and how instrument tags are built.

The first letter of a tag says what is being measured (P = pressure, F = flow, T = temperature, L = level). The second letter says what the instrument does (T = transmitter, I = indicator, C = controller). So “FIC-201” means a flow controller on loop 201. Here is what a real, small P&ID looks like once a control loop is added:

Example P&ID: Cross-Flow UF Skid with Pressure-Control Loop

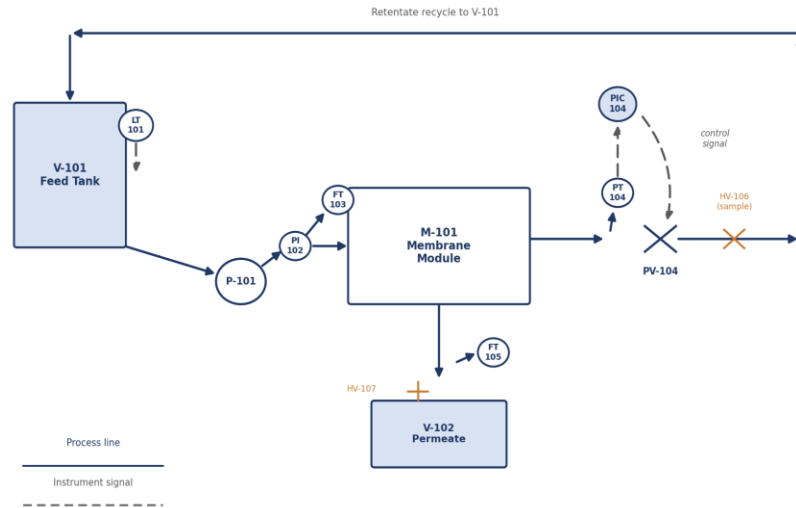


Figure 3.3 — An example P&ID: a pressure sensor (PT-104) feeds a controller (PIC-104), which adjusts a valve (PV-104). Solid lines are pipes; dashed lines are instrument signals — that difference is the single most important thing to notice on any real P&ID.

3.3 What a Pilot Plant Needs on Paper

A pilot that's meant to support a real scale-up decision needs more than just a drawing on the wall. At minimum, it needs:

Document	Why It Matters
Process description / design basis	The plain-language reference everything else gets checked against.
PFD + stream table	The big picture and the numbers (Section 3.1).
P&ID(s)	Full piping, valve, and instrument detail (Section 3.2).
Equipment list	Make, model, and materials for everything — needed for spares and for scale-up sizing.
Instrument list	Every tag, its range, and where its signal goes.
Line list	Every pipe: its service, size, and pressure/temperature rating.
SOPs	Step-by-step instructions for running, starting, and cleaning (Section 4.2).
Safety paperwork	Risk assessment, chemical safety sheets, and over-pressure protection.

Table 3.1 — The minimum paper trail a pilot needs.

This is the package that gets handed to a scale-up team, an investor doing due diligence, or a regulator. It's much cheaper to build this up as you go than to reconstruct it later from memory and old notebooks.

3.4 Real Example: Carlsbad Desalination Plant

The Carlsbad Desalination Plant in California is one of the most publicly documented large membrane plants anywhere — useful here even though its detailed P&IDs aren't public, for the same security reasons most working plants keep theirs private. Poseidon Water built it with IDE Technologies as the process partner, finishing in December 2015 at a cost of about \$592 million.

At the PFD level, the process is exactly the block sequence this guide describes: seawater intake, pretreatment, high-pressure reverse osmosis with energy-recovery devices (because pushing past seawater's natural osmotic pressure is the plant's single biggest energy cost), then remineralization and storage. The plant takes in up to 108 million gallons a day of seawater and produces up to 50–54 million gallons a day of drinking water — about 50% recovery — supplying roughly 10% of San Diego County's water.

A project at this scale needs the full documentation package from Table 3.1 — the operator's 30-year supply contract with the water authority is a performance guarantee, and a guarantee that size is only enforceable against a fully documented design and a tested instrument package.

4. Scale-Up & Commercialization

How to move from pilot to commercial scale without the project falling apart along the way.

4.1 Scaling Up Step by Step

Membrane processes scale up in stages, not in one big jump. The single most common cause of a failed scale-up is trying to leap more than 10–100× in throughput in one step — fouling and flow behaviour rarely scale in a simple straight line.

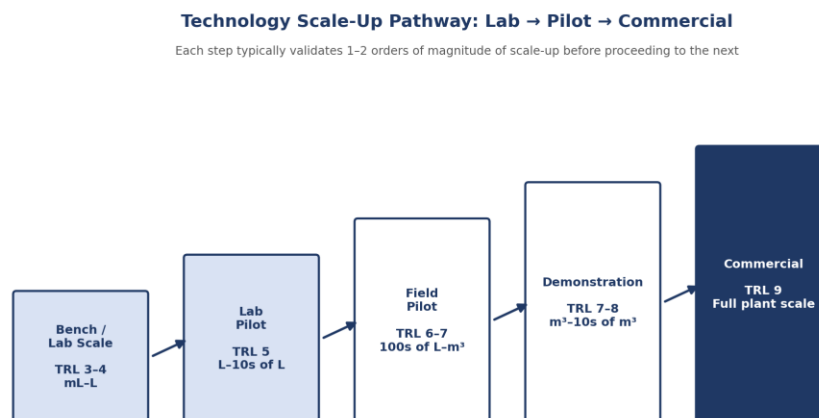


Figure 4.1 — A typical scale-up path, with rough Technology Readiness Levels (TRL) at each step.

There are two ways to scale up, and most real projects use both. “Scale-up by similarity” means keeping key numbers (like flow speed) the same at each size. “Scale-up by replication” means simply adding more identical membrane modules side by side, instead of building one bigger one — this is the standard, lower-risk approach for most membrane plants.

4.2 Writing SOPs

An SOP (Standard Operating Procedure) turns “what we did in the lab” into something anyone else can repeat correctly. A good SOP includes:

8. What it covers, and what it doesn't.
9. Safety steps and protective gear for that specific task.
10. The equipment and materials needed.
11. Numbered steps with clear targets (“raise to 4 bar $\pm$ 0.2 bar,” not “raise the pressure a bit”).
12. What data to record, and where.
13. A version history, since procedures keep improving over time.

4.3 Quality Control and Batch Records

As a process heads toward commercial use, “it worked that one time” has to become “it reliably hits spec, and we can prove it.” That means:

- A written spec for every important product quality, with a defined test method and limit.
- Batch records that store process settings and test results together, so a bad result can be traced back to its cause.
- A consistent sampling plan, so results from different runs can actually be compared.
- A change-control process, so any change to equipment or procedure is deliberate and written down.

4.4 Validating a Pilot (DQ / IQ / OQ / PQ)

This four-step framework, borrowed from regulated industries, is a useful checklist for de-risking any serious pilot:

Step	What It Proves
Design Qualification (DQ)	The planned design will actually meet the requirements — checked before anything is built.
Installation Qualification (IQ)	The equipment was installed correctly, matching the approved design.

Step	What It Proves
Operational Qualification (OQ)	The system works correctly across its full intended range, even without making real product.
Performance Qualification (PQ)	The system reliably makes product that meets spec, run after run.

Table 4.1 — The DQ/IQ/OQ/PQ sequence.

4.5 Thinking About the Economics

A technology can work perfectly and still fail commercially if the numbers don't add up. Even a rough early cost estimate should separate:

- CAPEX — equipment, installation, engineering, and a healthy contingency (often 15–30% this early, simply because so much is still uncertain).
- OPEX — energy (usually the single biggest line for pressure-driven membranes), membrane replacement, cleaning chemicals, labour, and waste disposal.
- Membrane replacement cost specifically — this depends on real lifetime under your real fouling and cleaning routine, not the manufacturer's generic warranty number.
- Sensitivity — show how the numbers change if your most uncertain guess (often membrane life) turns out to be wrong. If your cost estimate can't show this, it isn't finished yet.

4.6 How Ready Is the Technology?

The Technology Readiness Level (TRL) scale, from 1 to 9, gives everyone a shared way to talk about how far along a process is, and what's still needed before the next big investment:

TRL	Stage	What's Usually Needed Next
1–3	Basic research	Bench data showing it's technically possible.
4–5	Lab / small pilot	Repeatable results across the feed range you'll actually see.
6–7	Field pilot / demo	Continuous running at real scale, with real feed, fully documented (Section 3).
8	Fully qualified system	PQ-level proof (Section 4.4) and a believable cost estimate (Section 4.5).
9	Commercial	A reference plant actually running at full scale.

Table 4.2 — The TRL scale and what's typically needed to move up a level.

Being “ready” means more than a high TRL number. It also means the quality spec, the SOPs, and a reliable supply of membranes and spare parts are all in place. A technically great process with no quality spec and no backup membrane supplier still isn't ready for a real customer.

4.7 Real Example: E3 Lithium's Clearwater Project

Most scale-up stories are written years after the fact. This one is happening right now, which makes it a good way to see the TRL ladder mapped onto real, dated steps instead of a hypothetical. E3 Lithium is pulling battery-grade lithium straight out of deep salty water (brine) in central Alberta, instead of mining rock or using evaporation ponds, aiming for a commercial plant near Olds called the Clearwater Project.

The company's own announced timeline tracks Figure 4.1 almost exactly: a 2023 field pilot showed over 94% lithium recovery (roughly TRL 5–6); a demonstration plant near Olds started up in September 2025 and began making real battery-grade lithium carbonate (TRL 6); a second phase of that plant started up in June 2026; and a third phase — adding a larger, commercial-scale unit just to prove it works at that size — was approved in March 2026 (TRL 7). The full commercial plant aims for its first, smaller production train (about 12,000 tonnes/year) in 2028–2029, scaling up to 36,000 tonnes/year at a total cost of roughly USD \$2.46 billion (TRL 8–9).

Two things make this a useful teaching example. First, it's a clean real-world case of “scale-up by replication” (Section 4.1): E3 keeps duplicating the same modular equipment rather than jumping straight to one giant unit. Second, it shows why a cost estimate (Section 4.5) needs to separate the long-run resource (the company says the brine could support production for about 50 years) from the much narrower, nearer-term question of whether the first commercial module gets built on schedule.

5. Suggested Learning Path

If you're working through this guide on your own, this order builds each idea on the last one:

14. Start with Sections 1.1–1.2 to get the basic vocabulary down first.
15. Do the mass-balance equations in Section 1.4 by hand, once, on a simple example — before you trust a spreadsheet to do it for you.
16. Read Sections 1.5 and 2 together: sketch an operating window on paper, then see how it actually gets achieved and watched on real equipment.
17. Practice reading a few real PFDs and P&IDs (from any textbook or vendor manual) using Section 3 as your key, before trying to draw your own.
18. Come back to Section 4 every time a project is about to move up a TRL level — it's meant to be reused, not just read once.

6. Further Reading & Standards

This guide summarizes common engineering practice, not new research. For more depth, these are good next steps:

- Baker, R. W. *Membrane Technology and Applications*, 3rd ed.; Wiley — the standard general membrane textbook.
- Cheryan, M. *Ultrafiltration and Microfiltration Handbook*, 2nd ed.; CRC Press — practical UF/MF design, including the mass-balance and diafiltration equations used here.
- Mulder, M. *Basic Principles of Membrane Technology*; Springer — the theory behind membrane transport and module shapes.
- ISA-5.1-2009, *Instrumentation Symbols and Identification* — the standard behind the P&ID symbols and tags in Section 3.2.
- ISPE Baseline Guides (Commissioning and Qualification) — the industry-standard version of the DQ/IQ/OQ/PQ framework in Section 4.4.
- Towler, G.; Sinnott, R. *Chemical Engineering Design*, 2nd ed.; Butterworth-Heinemann — PFD/P&ID conventions and cost-estimating in a wider engineering context.

If you're working with organic-solvent-stable membranes specifically (for example, NMP/DMF-resistant OSN systems), pair this guide with material-specific stability data — the operating window in Section 1.5 is only as good as the membrane chemistry behind it.

Sources for the Worked Examples

- MAX-DEWAX® (Section 1.6): Oil & Gas Journal, “New membrane process debottlenecks solvent dewaxing unit”; Marchetti et al., *Chem. Rev.* 2014, 114, 10735–10806; Buonomenna, *RSC Green Chem.* 2014 (OSN energy/solvent-recovery figures).
- Whey protein UF (Section 2.6): *Int. J. Dairy Technol.* multistage UF/MF reviews; ScienceDirect, “Whey protein concentrate production by continuous ultrafiltration” (12-stage industrial plant configuration).
- Carlsbad Desalination Plant (Section 3.4): San Diego County Water Authority and Water Collaborative Delivery Association project profiles; IDE Technologies company materials.
- E3 Lithium Clearwater Project (Section 4.7): E3 Lithium Ltd. news releases and corporate disclosures (2025–2026), including Demonstration Facility Phase 2/3 updates reported via Resource World Magazine and BNN Bloomberg.